Notifications Feature — Functional BRD

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This BRD defines what the Meridian trading platform must do to tell customers about trades, price moves, account events and security events, on web, mobile and email. It is written against the Meridian prototype (`trading-platform.html`) and is ready for business review.

Document control

Field	Value
Document	Notifications Feature — Functional BRD
Product	Meridian trading platform (web terminal + mobile app)
Version	0.1 (first draft)
Status	Draft for review
Business owner	Product Manager, Trading Experience
Reviewers	Engineering lead, UX lead, QA lead, Compliance, Security
Approver	Head of Product

Version	Change	Author
0.1	First draft from the Meridian prototype	Product

Purpose and objectives

Customers must learn about fills, triggered alerts and security events within seconds, on the channel they chose, without being flooded. Today the prototype shows only in-app toasts and a bell list that are lost on page reload, with no email, push, preferences or delivery tracking.

#	Business objective	Success metric	Target
O1	Customers act on market moves in time	Median time from trigger to delivery (push / in-app)	⩽ 5 s
O2	Fewer “was my order filled?” support contacts	Order-status contacts per 1,000 active traders	−30% in 6 months
O3	Higher engagement with price alerts	Share of active traders with ⩾ 1 active alert	35%
O4	Faster response to account takeover attempts	Share of new-device sign-ins confirmed or blocked within 15 min	⩾ 80%
O5	Avoid notification fatigue	Push opt-out rate per month	⩽ 2%

Baselines for O2–O5 are not yet measured; see Open questions.

Scope

Release 1 covers four channels (in-app, mobile push, email, SMS for security only) and the event types in the catalogue below.

In scope

- In-app notification centre (bell, list, unread count, filters, mark as read)
- Real-time toasts while the user is on the platform
- Mobile push (iOS, Android) and email; SMS for security and one-time-code events only
- Customer-created price alerts (crosses above / below, day change %) and their lifecycle
- System notifications: order placed / filled / partially filled / cancelled / rejected, take-profit and stop-loss hits
- Account and security notifications: new device sign-in, password change, deposits and withdrawals
- Notification preferences per category and channel, quiet hours, time zone
- Delivery tracking, retries, fallback and an audit log

Out of scope (later releases)

- Indicator-based alerts (e.g. RSI crosses 70) and drawing-based alerts (trend-line touch)
- Marketing campaigns and promotional messages

- Webhooks, Slack, Telegram and other third-party channels
- Price alerts on symbols the customer cannot trade in their region

Assumptions

- Real-time market data and order events are available on an internal event stream.
- Customers have a verified email; push needs app install and OS permission.
- A third-party provider is used per channel (e.g. APNs/FCM for push, an email and an SMS gateway).

Dependencies

- Order management system (order and fill events)
- Market data service (last price, previous close)
- Identity service (sign-in, device and session events)
- Payments / funding service (deposits, withdrawals)
- Compliance review of trade-confirmation wording

Constraints

- Trade confirmations must meet regulatory record-keeping rules for the operating jurisdictions.
- Notification text must never include a full account number, password or one-time code beyond the code itself.

Stakeholders and personas

Stakeholder	Interest	Role in this project
Product Manager, Trading Experience	Engagement, retention	Owns requirements, accepts stories
Compliance	Trade confirmations, record keeping, wording	Approves templates and retention rules
Security / Fraud	Account takeover response	Defines security events and their channels
Customer Support	Fewer “where is my order” contacts	Uses the delivery log to answer customers
Engineering (platform, mobile)	Build and run the service	Estimates, designs, delivers

Stakeholder	Interest	Role in this project
QA	Testable acceptance criteria	Writes the test plan from this BRD

Persona	Who they are	What they need from notifications
Active day trader	Trades several times a day on the web terminal	Instant fill and alert toasts; no email noise; sound optional
Long-term investor	Checks the mobile app a few times a week	A few push messages on big moves; weekly email digest
New customer	Just funded the account	Clear deposit confirmation and guidance; easy controls to turn things off
Support agent (internal)	Answers customer contacts	See what was sent, when, on which channel, and whether it was delivered

Notification catalogue

Release 1 has 16 event types in four categories. Priority sets delivery speed and whether quiet hours apply: **Critical** always delivers immediately; **High** delivers immediately but respects quiet hours for push; **Normal** can be batched. “Locked” means the customer cannot turn that channel off.

ID	Category	Event	Trigger	Priority	Default channels	Example message
N-01	Trading	Order placed	Limit or stop order accepted by the OMS	Normal	In-app	Limit buy 10 AAPL @ 222.00 placed
N-02	Trading	Order filled	Order fully executed	High	In-app, push, email (locked)	Bought 10 AAPL @ 222.00
N-03	Trading	Order partially filled	Part of an order executed	High	In-app, push	4 of 10 AAPL filled @ 222.00
N-04	Trading	Order cancelled	Customer or system cancels an order	Normal	In-app	Limit buy AAPL cancelled
N-05	Trading	Order rejected	OMS rejects an order	High	In-app, push	Order rejected: insufficient buying power

ID	Category	Event	Trigger	Priority	Default channels	Example message
N-06	Trading	Order expired	Day order reaches end of session unfilled	Normal	In-app, email	Day order for AAPL expired
N-07	Trading	Take profit hit	Price reaches the position's take-profit level	High	In-app, push	Take profit hit on AAPL @ 237.67
N-08	Trading	Stop loss hit	Price reaches the position's stop-loss level	High	In-app, push	Stop loss hit on TSLA @ 275.00
N-09	Price alert	Price crosses above	Last price moves from below to at/above the level	High	In-app, push	AAPL crossed above 232.00 — now 232.14
N-10	Price alert	Price crosses below	Last price moves from above to at/below the level	High	In-app, push	BTC-USD crossed below 62,000
N-11	Price alert	Day change exceeds %	Change vs previous close reaches ±X%	High	In-app, push	NVDA is up 5.2% today
N-12	Account	Deposit received	Funds credited	Normal	In-app, email	$5,000.00 deposit received
N-13	Account	Withdrawal requested / sent	Withdrawal created or paid out	High	In-app, email (locked)	$1,200.00 withdrawal sent to ••4821
N-14	Account	Margin / buying-power warning	Buying power falls below a set threshold	Critical	In-app, push, email (locked)	Buying power is below $500
N-15	Security	New device sign-in	Sign-in from an unrecognised device	Critical	Push, email, SMS (locked)	New sign-in from Chrome on Windows, Denver
N-16	Security	Password or 2FA changed	Credential change completed	Critical	Email, SMS (locked)	Your password was changed

Functional requirements

There are 30 requirements in eight areas: 21 Must, 7 Should and 2 Could. Each has acceptance criteria that QA can test directly.

A. In-app notification centre

ID	Requirement	Priority	Acceptance criteria
FR-A1	Show a bell icon in the header with an unread count.	Must	Count matches unread items; hidden at 0; shows “99+” above 99; updates within 2 s of a new item.
FR-A2	Opening the bell shows a panel of the 50 most recent notifications, newest first.	Must	Each item shows icon, title, one-line body, relative time and unread dot; “View all” opens the full page.
FR-A3	A full notifications page lists history for 90 days with filters by category and read state.	Should	Filters combine; empty state shown when nothing matches; paging or infinite scroll beyond 50 items.
FR-A4	Users can mark one or all notifications as read.	Must	Read state syncs across web and mobile within 5 s; count updates immediately.
FR-A5	Each notification deep-links to its subject (order, position, alert, chart, security page).	Must	Click opens the correct screen with the item in focus; a deleted subject shows a clear “no longer available” message.
FR-A6	Security notifications offer inline actions: “Yes, it was me” and “No, secure my account”.	Must	“No” signs out other sessions and starts password reset; both actions are audit-logged.

B. Real-time toasts

ID	Requirement	Priority	Acceptance criteria
FR-B1	Show a toast for High and Critical events while the user is signed in on web.	Must	Toast appears within 2 s of the event; auto-dismisses after 6 s; Critical toasts stay until dismissed.
FR-B2	Toasts stack (max 3 visible) and never cover the order ticket’s submit button.	Should	A 4th toast queues; layout verified at 1280 px and 390 px widths.
FR-B3	Toasts are announced to screen readers.	Must	Container uses a polite live region; Critical uses assertive.

ID	Requirement	Priority	Acceptance criteria
FR-B4	Optional sound for fills and alerts.	Could	Off by default; toggle in preferences; respects quiet hours.

C. Price alerts

ID	Requirement	Priority	Acceptance criteria
FR-C1	Create an alert from the chart toolbar, the Alerts page or a symbol's page.	Must	Form pre-fills symbol and a level 2% from last price; shows distance from current price as you type.
FR-C2	Conditions: crosses above, crosses below, day change ≥ +X%, day change ≤ −X%.	Must	Each condition fires exactly once per crossing; a level already passed at creation shows a warning.
FR-C3	Alert levels show as draggable lines on the chart.	Should	Dragging updates the level and confirms with a toast; the change is saved.
FR-C4	Pause, resume, edit and delete alerts.	Must	Paused alerts never fire; resuming re-arms from the current price.
FR-C5	Expiry: alerts can be one-time (default) or recurring, and expire after a chosen date (default 90 days).	Should	One-time alerts move to "Triggered"; expired alerts move to "Expired" and notify once.
FR-C6	Limit of 200 active alerts per customer.	Must	The 201st shows an inline error naming the limit.

D. Preferences

ID	Requirement	Priority	Acceptance criteria
FR-D1	A preferences matrix of category × channel with toggles.	Must	Locked cells are visible but disabled, with a tooltip explaining why.

ID	Requirement	Priority	Acceptance criteria
FR-D2	Quiet hours with start, end and time zone.	Should	During quiet hours, Normal and High push is held and delivered as one summary at the end; Critical still delivers.
FR-D3	Changes apply immediately and sync across devices.	Must	A message sent 5 s after a change follows the new setting.
FR-D4	One-click unsubscribe link in every non-locked email.	Must	Link turns off that category's email without sign-in; confirmation page shown.

E. Delivery channels

ID	Requirement	Priority	Acceptance criteria
FR-E1	Deliver push via APNs (iOS) and FCM (Android) to every registered device.	Must	Invalid tokens are removed after the provider reports them; signed-out devices receive nothing.
FR-E2	Deliver email with branded HTML and a plain-text part.	Must	Renders in the top 5 email clients; trade confirmations include order ID, time (UTC and local), quantity, price, fees.
FR-E3	SMS only for Security events and one-time codes.	Must	Message ⩽ 160 characters; no links in security SMS.
FR-E4	Fallback: if push fails or the device is unreachable for 5 min, send a Critical event by email, then SMS.	Should	Fallback path is recorded in the delivery log.
FR-E5	Weekly email digest of alerts triggered and orders filled.	Could	Sent Monday 08:00 local time; skipped when there was no activity.

F. Content and templates

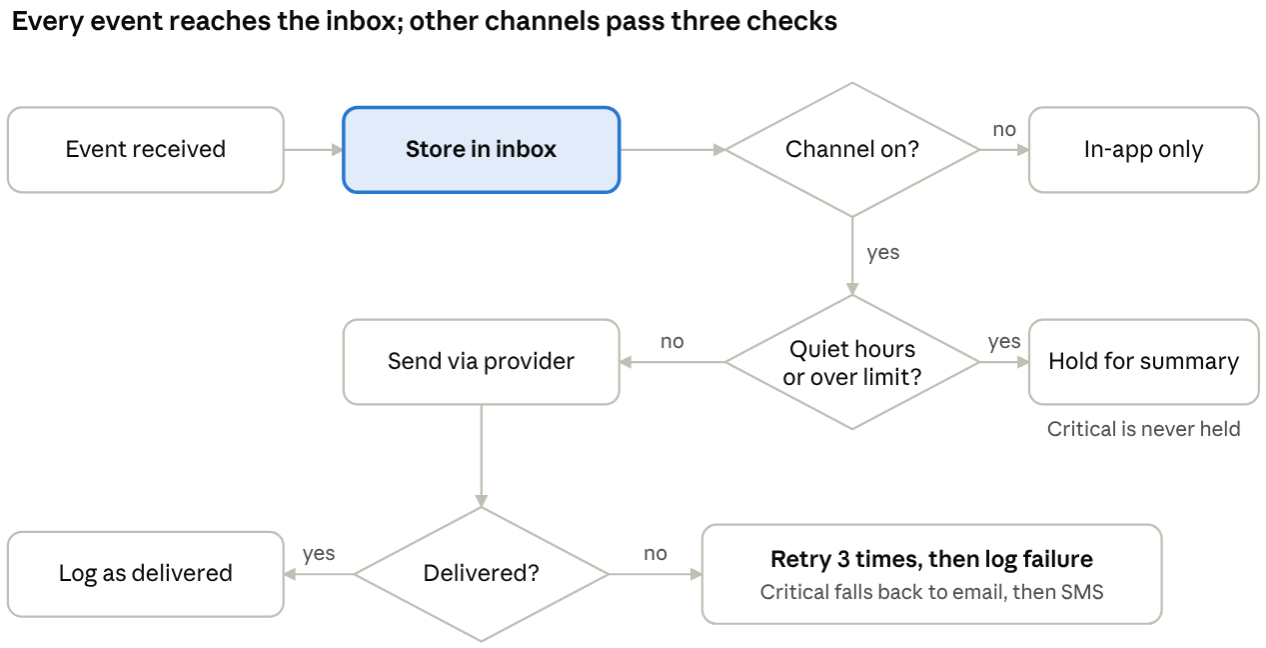
ID	Requirement	Priority	Acceptance criteria
FR-F1	Every event type has a template per channel with variables (symbol, qty, price, time).	Must	Missing variables block sending and raise an error, never a message with blank fields.

ID	Requirement	Priority	Acceptance criteria
FR-F2	Prices use the symbol's decimal places; amounts use the account currency; times use the user's time zone.	Must	EUR/USD shows 5 decimals; BTC quantity up to 4 decimals.
FR-F3	Templates support localisation.	Should	English at launch; strings externalised for later languages.

G. Support and audit

ID	Requirement	Priority	Acceptance criteria
FR-G1	Support agents can search a customer's notifications and see each channel's delivery status.	Must	Statuses: queued, sent, delivered, opened, failed (with reason), suppressed (with reason).
FR-G2	Admins can pause a single event type platform-wide during an incident.	Must	Pause and resume are audit-logged with the admin's ID and reason.

User and system flows



delivery flow · one event, per channel

The inbox always gets the notification. Each other channel is checked on its own: preference (or lock), then quiet hours and throttling, then delivery with retries.

Create a price alert

1. Customer clicks Alert on the chart toolbar, or New alert on the Alerts page.
2. Form opens with the symbol, “Crosses above” and a level 2% above the last price pre-filled.
3. Customer edits condition, level, expiry and channels; the form shows the distance from the current price.
4. Customer clicks Create alert; the alert appears in the list as Active and as a line on the chart.
5. When the condition is met, the system sends N-09, N-10 or N-11 and sets a one-time alert to Triggered.

Read and act on notifications

1. Bell shows the unread count; customer opens the panel.
2. Customer clicks an item; it is marked read and the linked screen opens.
3. For a security item, the customer picks “Yes, it was me” or “No, secure my account” directly in the panel.
4. “Mark all as read” clears the count on every device.

Change preferences

1. Customer opens Profile → Notifications.
2. Customer turns channels on or off per category; locked cells cannot be changed and explain why.
3. Customer sets quiet hours and time zone.
4. Changes save immediately and a confirmation toast appears.

Business rules

These rules decide whether, when and how often a message goes out. They apply in the order listed.

ID	Rule	Detail
BR-01	Locked channels	Trade confirmations (N-02) by email, withdrawals (N-13), buying-power warnings (N-14) and all Security events cannot be turned off on their locked channels.

ID	Rule	Detail
BR-02	Preferences	For every other event and channel, the customer's preference decides. New customers get the default channels in the catalogue.
BR-03	Quiet hours	Default 22:00–07:00 local, off until the customer turns it on. Held push messages are sent as one summary when quiet hours end. Critical events ignore quiet hours.
BR-04	De-duplication	The same event ID is never delivered twice on the same channel. A partial fill followed by a full fill within 2 s sends only the full fill.
BR-05	Throttling	At most 20 push messages per customer per hour. Beyond that, Normal and High events go to the in-app centre only, plus one push saying how many were held. Critical is never throttled.
BR-06	Price-alert firing	Alerts fire on the last traded price, evaluated on every tick. A crossing alert needs the previous tick on the other side of the level. One-time alerts disarm after firing.
BR-07	Market hours	Stock alerts fire only during regular and extended sessions. Crypto alerts fire 24/7. FX alerts fire Sunday 17:00 to Friday 17:00 New York time.
BR-08	Retries	Push and email: retry 3 times at 30 s, 2 min and 10 min. SMS: retry once after 60 s. Then mark as failed and apply fallback (FR-E4) if Critical.
BR-09	Expiry of stale messages	A High price-alert push not sent within 10 minutes of its trigger is dropped from push (still shown in-app), because the price may no longer apply.
BR-10	Sensitive data	Messages may show the last 4 digits of an account, never the full number. Security SMS contain no links.
BR-11	Signed-out devices	Push goes only to devices with an active session. Signing out removes the device token.
BR-12	Record keeping	Trade confirmations and their delivery records are kept for the regulatory retention period (see Data requirements).

Data requirements

Five records support the feature. A notification is created once per event and has one delivery record per channel.

Entity	Key fields	Retention
Notification	notification_id, customer_id, event_type (N-xx), event_id, priority, title, body, deep_link, created_at, read_at	90 days in the app; trade confirmations per regulatory rule (assumed 7 years)
Delivery	delivery_id, notification_id, channel, provider_message_id, status, failure_reason, attempts, sent_at, delivered_at, opened_at	Same as its notification
Price alert	alert_id, customer_id, symbol, condition, value, recurring, expires_at, status (Active, Paused, Triggered, Expired), last_triggered_at	Until deleted, plus 90 days
Preference	customer_id, category, channel, enabled, quiet_hours_start, quiet_hours_end, time_zone, updated_at	While the account is open
Device	device_id, customer_id, platform, push_token, last_seen_at, session_active	Removed on sign-out or after 90 days unused

- Every preference change, alert change, admin pause and security inline action is written to the audit log with who, what, when and from which device.
- Customers can export their notification history (CSV) from the full notifications page.
- Personal data in notifications follows the platform's data-protection policy; deleting an account deletes notifications except records kept for regulatory reasons.

Non-functional requirements

ID	Area	Requirement
NFR-01	Latency	Event to in-app and push hand-off: p50 ⩽ 2 s, p95 ⩽ 5 s. Email hand-off p95 ⩽ 60 s.

ID	Area	Requirement
NFR-02	Throughput	50,000 notifications per minute sustained; 5× burst for 10 minutes at market open and on large market moves.
NFR-03	Alert evaluation	1 million active price alerts evaluated on every tick without adding more than 1 s to NFR-01.
NFR-04	Availability	99.9% monthly for the notification service; in-app centre readable even when channel providers are down.
NFR-05	Durability	No accepted event is lost: events are persisted before acknowledgement and replayed after an outage.
NFR-06	Security	Encryption in transit (TLS 1.2+) and at rest; provider credentials in a secrets manager; signed deep links.
NFR-07	Privacy	Lock-screen push for Security and account events hides amounts unless the customer opts in.
NFR-08	Accessibility	WCAG 2.2 AA: keyboard reachable bell and list, live-region announcements, contrast ⩾ 4.5:1, status never shown by colour alone.
NFR-09	Observability	Dashboards for volume, latency, failure rate by channel and provider; alert on failure rate > 2% for 5 min.
NFR-10	Compatibility	Latest 2 versions of Chrome, Safari, Edge, Firefox; iOS 16+; Android 10+.

Reporting and analytics

- Daily volume by event type and channel; delivery, open and failure rates by channel and provider.
- Trigger-to-delivery latency (p50, p95) for fills and price alerts, tracked against O1.
- Opt-out rate by category and channel (O5); alerts created, triggered and deleted per active trader (O3).
- Security inline actions: share confirmed, share secured, time to response (O4).

Risks

Risk	Impact	Mitigation
Burst of alerts at market open delays fills	Late trade confirmations	Separate queues per priority; Critical and fills first
Push provider outage	Customers miss alerts	In-app still works; email/SMS fallback for Critical (FR-E4)
Notification fatigue	Opt-outs, app uninstalls	Throttling (BR-05), quiet hours, digest, sensible defaults
Stale price alerts delivered late	Customer trades on old prices	Drop push after 10 min (BR-09); price and time in every message
Wording not approved by Compliance	Launch delay	Template review starts in sprint 1

Open questions

- [ ] Which jurisdictions launch first, and what is each one's retention period for trade confirmations?
- [ ] Is SMS needed at launch, or can security rely on push and email?
- [ ] Which email, SMS and push providers will we use?
- [ ] What are today's baselines for O2–O5?
- [ ] Should price alerts fire on bid/ask instead of last price for FX and crypto?

Sign-off

Role	Decision	Date
Head of Product		
Engineering lead		
Compliance		
Security		
QA lead		